

Junyi (Conny) Zhou

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EDUCATION

- Harvard University** Boston, MA
M.S. in Health Data Science August 2025 – March 2027
 - Program requirements complete December 2026 (commencement March 2027); available for full-time start January 2027
 - Coursework: Deep Learning, AI in Medicine, Healthcare Machine Learning, Data Science
- Emory University, College of Arts and Sciences** Atlanta, GA
B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 August 2021 – May 2025
 - Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures, Database Systems

TECHNICAL SKILLS

- Languages:** Python, C++, Java, SQL, R, JavaScript
- ML & Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers
- LLM & Applied AI:** RAG, agentic pipelines, prompt engineering, embeddings & vector search, OpenAI API, MongoDB Atlas Vector Search
- Data & Compute:** PostgreSQL, MySQL, MongoDB, Spark, Hadoop, SLURM/HPC, multi-GPU job scheduling
- Backend & DevOps:** Flask, REST APIs, React, Docker, Auth0, Git
- Cloud:** AWS (SageMaker, Lambda, EC2, S3, Redshift, Elastic Beanstalk, CloudFront), Snowflake, BigQuery
- Certifications:** AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

MACHINE LEARNING PROJECTS

- Parameter-Efficient Fine-Tuning of BLIP-2 for Visual Question Answering** [GitHub] Boston, MA
Harvard SHBT-261 Spring 2026
 - Fine-tuned **BLIP-2** (OPT-2.7B) on **TextVQA** using **LoRA**, lifting accuracy from 0.078 zero-shot to 0.213 — nearly $3\times$ — while training 2.62M parameters (0.07% of the model) in 8 minutes on a single **A100**.
 - Isolated the contribution of **prompt engineering** across four templates, showing OCR-augmented prompting alone raised zero-shot accuracy to 0.106 before any weight updates.
 - Ablated **LoRA** rank over $r \in \{4, 8, 16\}$ and automated the pipeline end to end, generating every reported figure and **L^AT_EX** table directly from result JSONs for reproducibility.
- Airway Management Simulation Chatbot** [Live Demo] Atlanta, GA
Full-Stack Engineer, Scrum Master — Emory Pediatric Hospital January 2024 – August 2025
 - Architected a clinical **agentic LLM pipeline** pairing **RAG** retrieval over **MongoDB Atlas Vector Search** with **prompt engineering** and context-window management to generate multi-turn training scenarios from **OpenAI** models.
 - Built a **HIPAA-conscious** full-stack platform (**Python/Flask** REST API, React, MongoDB, **Auth0**) deployed on **AWS** Elastic Beanstalk and CloudFront, tracking resident performance metrics.
 - Led an Agile team of 6, running code reviews and folding clinician feedback into successive releases.

RESEARCH EXPERIENCE

- Cross-Species Drug Translation using Optimal Transport & Generative AI** Boston, MA
Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute February 2026 – Present
 - Built **Optimal Transport** models (**VAEs**, **unbalanced OT**) predicting animal-to-human trial translation from **scRNA-seq** atlases, with **scanpy** pipelines automating cell-type labeling and dataset integration.
 - Ported a 2018 CPU-only **TensorFlow** reference implementation to **PyTorch**, reimplementing the **Input Convex Neural Networks** behind the optimal-transport dual and enabling GPU training on Harvard's **FASRC Cannon** cluster.
 - Benchmarked **CellOT** against **scGen** on cross-species perturbation prediction, measuring 0.85–0.90 R^2 with the source species in training versus 0.65–0.67 on the held-out-species split.
 - Established through **paired single-factor ablations** that gene-space choice reverses with split — 6,619 orthologs wins in-distribution while 1,000 HVGs wins on every out-of-distribution metric — ruling out a split-independent default.
- Autonomous Ablation-Search Agent for Cluster-Scale ML Experimentation** [GitHub] Boston, MA
Independent Project — Harvard FASRC Cannon Cluster June 2026 – Present
 - Built a **self-directed experiment harness** (submit → watch → reflect → decide → synthesize) that plans, launches, and triages **SLURM** ablation studies unattended — **338 runs**, 318 clean and 20 with handled errors, over **79 hours** of compute.
 - Split the planner from the execution substrate and **checkpointed each agenda to disk**, so the cluster keeps running the last plan while the planner is offline; an adaptive one-delta proposer clears a **validation gate** before every submit.
 - Designed a **fairshare-aware scheduler** routing small-MLP training to CPU partitions, where GPU carries 200–500× the fairshare cost, and made runs **preemption-tolerant** on `serial_requeue` via checkpoint/resume and capped job arrays.
 - Wired an optional **LLM planner** behind that interface with per-call **token and cost accounting** and a spend ceiling that degrades to the deterministic planner on breach or API failure.